ECA1804 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO1 ENGINEERING DESIGN ASSIGNMENT

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Introduction

This assignment considers the design of a 60 km optical-fibre communication link operating at a wavelength of 1550 nm. The initial data rate is 10 Gb/s, and the design should consider a possible future upgrade to 100 Gb/s. The main design checks are fibre selection, received optical power, dispersion, polarization-mode dispersion (PMD), optical signal-to-noise ratio (OSNR), and final deployment suitability.

Stage 1 – Interpret the Engineering Requirement

The link must carry information over 60 km at 1550 nm. The first design rate is 10 Gb/s. A suitable design should also allow a practical upgrade path toward 100 Gb/s.

1.1 Important terms

Optical fibre: A thin glass waveguide that carries information as pulses of light.

Wavelength (λ): The optical wavelength. Here $\lambda = 1550$ nm $= 1.55$ μm.

Attenuation: Reduction in optical power as light travels through fibre, normally measured in dB/km.

Dispersion: Spreading of optical pulses as they travel, which can cause neighbouring bits to overlap.

Receiver sensitivity: The minimum received optical power needed for the receiver to operate to its specified performance.

OSNR: The ratio of optical signal power to optical noise power, measured in a stated reference bandwidth.

1.2 Design targets

Distance: 60 km

Operating wavelength: 1550 nm

Initial data rate: 10 Gb/s

Possible future data rate: 100 Gb/s

Transmitter power: +3 dBm

Receiver sensitivity: −20 dBm

Required power margin: at least 6 dB

Required OSNR: 17 dB in the stated reference bandwidth

Stage 2 – Select the Fibre and Propagation Configuration

A single-mode fibre is generally preferred for a 60 km telecom link because it avoids the large intermodal dispersion associated with multimode fibre.

2.1 Numerical aperture (NA)

Formula: $NA = \sqrt{(n_1^2 - n_2^2)}$

Fibre A: $n_1 = 1.4500$, $n_2 = 1.4450$

$NA = \sqrt{(1.4500^2 - 1.4450^2)} = \sqrt{(2.102500 - 2.088025)} = \sqrt{0.014475} \approx 0.1203$

Fibre B: $n_1 = 1.480$, $n_2 = 1.465$

$NA = \sqrt{(1.480^2 - 1.465^2)} = \sqrt{(2.190400 - 2.146225)} = \sqrt{0.044175} \approx 0.2102$

2.2 Normalised frequency (V-number)

Formula: $V = (2\pi a/\lambda) \times NA$, where a is the core radius and λ is the wavelength.

Use $\lambda = 1.55$ μm.

Fibre A: $a = 4.0$ μm, $NA = 0.1203$

$V = (2\pi \times 4.0 / 1.55) \times 0.1203 \approx 1.95$. Since $V < 2.405$, Fibre A is single-mode at this wavelength.

Fibre B: $a = 25$ μm, $NA = 0.2102$

$V = (2\pi \times 25 / 1.55) \times 0.2102 \approx 21.30$. Since $V > 2.405$, Fibre B is multimode.

Approximate step-index multimode count: $M \approx V^2/2 = 21.30^2/2 \approx 227$ modes. For comparison, an ideal parabolic graded-index fibre is often approximated by $M \approx V^2/4 \approx 113$ modes; this comparison does not mean Fibre B is graded-index.

2.3 Cut-off wavelength for Fibre A

Formula: $\lambda c = 2\pi aNA / 2.405$

$\lambda c = (2\pi \times 4.0 \times 0.1203) / 2.405 \approx 1.257\ \mu m = 1257\ nm$.

Because 1550 nm is above the calculated cut-off wavelength, Fibre A operates in the single-mode region at 1550 nm.

2.4 Fibre choice

Recommended starting point: use a standards-compliant single-mode fibre for the main span, such as G.652.D, and consider G.657.A bend-insensitive fibre at a building entry or tight-bend section. G.655 may be investigated as a lower-dispersion alternative, subject to checking the actual fibre specification, attenuation, compatibility, and availability.

I recommend single-mode fibre (G.655) with a low-dispersion propagation configuration because it reduces chromatic dispersion over the 60 km link, helps minimise pulse broadening, and supports the potential upgrade from 10 Gb/s to 100 Gb/s.

Stage 3 – Optical Power Budget

The power budget checks whether enough optical power reaches the receiver after all link losses.

3.1 Fibre attenuation

Given attenuation components: 0.04 + 0.11 + 0.03 = 0.18 dB/km.

Fibre loss = attenuation × distance = 0.18 dB/km × 60 km = 10.8 dB.

The fibre alone therefore causes 10.8 dB of loss over 60 km.

3.2 Total loss using G.652.D at the entry

Connector loss = 4 × 0.4 dB = 1.6 dB.

Splice loss = 10 × 0.08 dB = 0.8 dB.

Entry loss = 0.8 dB.

Total loss = 10.8 + 1.6 + 0.8 + 0.8 = 14.0 dB.

Received power = transmitter power − total loss = +3 dBm − 14.0 dB = −11.0 dBm.

Power margin = received power − receiver sensitivity = (−11.0) − (−20) = 9.0 dB.

The 9.0 dB margin exceeds the required 6 dB by 3.0 dB.

3.3 Total loss using G.657.A at the entry

Using an entry loss of 0.1 dB: total loss = 10.8 + 1.6 + 0.8 + 0.1 = 13.3 dB.

Received power = +3 − 13.3 = −10.3 dBm.

Power margin = (−10.3) − (−20) = 9.7 dB.

This option saves 14.0 − 13.3 = 0.7 dB compared with the G.652.D entry assumption and meets the 6 dB margin requirement.

3.4 Loss-reduction measures

Use appropriate low-loss connectors and splices.

Clean and inspect connector end faces before mating.

Avoid unnecessary connectors and tight bends.

Use bend-insensitive fibre where the route requires tight bends.

I recommend a power-budget configuration using single-mode fibre with G.657.A at the entry point because the total link loss is 13.3 dB, giving a received power of −10.3 dBm and a power margin of 9.7 dB, which exceeds the required 6 dB margin.

Stage 4 – Dispersion and PMD Management

4.1 Chromatic dispersion

Formula: $\Delta\tau = |D| \times \Delta\lambda \times L$, where D is dispersion in ps/(nm·km), $\Delta\lambda$ is source spectral width in nm, and L is fibre length in km.

Given $\Delta\lambda = 0.4$ nm and L = 60 km.

For G.652.D, D = 17 ps/(nm·km): $\Delta\tau = 17 \times 0.4 \times 60 = 408$ ps.

For G.655, D = 4.5 ps/(nm·km): $\Delta\tau = 4.5 \times 0.4 \times 60 = 108$ ps.

If the transmitter pulse width is 30 ps and broadening terms are combined by root-sum-square (RSS):

G.652.D: $\tau_{total} = \sqrt{(408^2 + 30^2)} \approx 409.1$ ps.

G.655: $\tau_{total} = \sqrt{(108^2 + 30^2)} \approx 112.1$ ps.

At 10 Gb/s, the bit period is $T_b = 1/R_b = 1/(10 \times 10^9) = 100$ ps. Both estimates exceed one bit period, so dispersion needs attention under this simplified model. Actual acceptance must use the system-specific dispersion tolerance and transceiver specifications.

4.2 Dispersion-compensating fibre (DCF)

For G.652.D, use Dmain = +17 ps/(nm·km), Lmain = 60 km, and DDCF = −100 ps/(nm·km).

Length required for approximate compensation: LDCF = −(Dmain × Lmain)/DDCF.

LDCF = −(17 × 60)/(−100) = 10.2 km.

If DCF attenuation is 0.5 dB/km, DCF loss = 10.2 × 0.5 = 5.1 dB.

Using the G.657.A entry total from Stage 3: new total loss = 13.3 + 5.1 = 18.4 dB.

Received power = +3 − 18.4 = −15.4 dBm. Margin = (−15.4) − (−20) = 4.6 dB.

This is below the required 6 dB margin by 1.4 dB. Therefore, this uncompensated power budget does not pass without redesign, additional suitable gain, or another dispersion strategy.

4.3 Polarization-mode dispersion (PMD)

Formula: PMD ≈ DPMD × √L.

Given DPMD = 0.1 ps/√km and L = 60 km: PMD = 0.1 × √60 ≈ 0.775 ps.

At 10 Gb/s, Tb = 100 ps and 0.1Tb = 10 ps. The estimated PMD of 0.775 ps is below 10 ps.

At 100 Gb/s, Tb = 10 ps and 0.1Tb = 1 ps. The estimate is below 1 ps but is close enough that the upgrade should be checked against the chosen transceiver and link specification.

I recommend a low-dispersion G.655 single-mode fibre with dispersion compensation and PMD monitoring because it reduces chromatic pulse broadening over the 60 km link, while the estimated PMD of 0.775 ps is within the simplified limits for 10 Gb/s and 100 Gb/s operation.

Stage 5 – OSNR and Amplification

OSNR is the optical signal-to-noise ratio in a specified reference bandwidth. In dB, it is calculated by subtracting the noise power from the signal power when both are measured in the same bandwidth.

5.1 Initial OSNR

Signal power = −16 dBm; ASE noise power = −39 dBm.

OSNR = −16 − (−39) = 23 dB.

Margin above the required OSNR = 23 − 17 = 6 dB. The initial OSNR meets the stated target.

5.2 OSNR after the amplifier

Assumed amplifier gain = 14 dB and ASE noise increase = 16 dB.

Amplified signal power = −16 + 14 = −2 dBm.

Amplified ASE noise power = −39 + 16 = −23 dBm.

OSNR after amplification = −2 − (−23) = 21 dB.

The OSNR decreases by 23 − 21 = 2 dB, but it remains 21 − 17 = 4 dB above the required target. Amplifier gain and noise figures must be confirmed from real equipment data; gain alone does not guarantee a valid link.

I recommend a low-noise optical amplifier with carefully controlled gain and OSNR monitoring because the calculated OSNR after amplification is 21 dB, which exceeds the required 17 dB by 4 dB, ensuring the link meets the specified OSNR target.

Stage 6 – Verification and Design Review

Check	Calculated result	Assessment
Power budget: G.652.D entry, no DCF	Margin = 9.0 dB	Passes 6 dB target
Power budget: G.657.A entry, no DCF	Margin = 9.7 dB	Passes 6 dB target
G.652.D plus 10.2 km DCF	Margin = 4.6 dB	Fails 6 dB target
Chromatic dispersion, 10 Gb/s	408 ps for G.652.D; 108 ps for G.655	Needs system-specific dispersion management
Estimated PMD	0.775 ps	Passes simplified limits; verify upgrade
OSNR before amplifier	23 dB	Passes 17 dB target
OSNR after amplifier	21 dB	Passes 17 dB target

Stage 7 – Proposed Integrated Design

A candidate design for further engineering verification is:

10 Gb/s optical transmitter operating at 1550 nm, launch power +3 dBm.

Low-loss connectors and splices with inspected and cleaned interfaces.

60 km single-mode main span; investigate G.655 where its lower dispersion is beneficial and compatible.

G.657.A bend-insensitive fibre at the building entry or tight-bend section.

7.1 Illustrative G.655 low-dispersion calculation

If G.655 dispersion is 4.5 ps/(nm·km), the approximate DCF length using DDCF = −100 ps/(nm·km) is:

LDCF = −(4.5 × 60)/(−100) = 2.7 km.

At 0.5 dB/km, DCF loss = 2.7 × 0.5 = 1.35 dB.

Assuming, only for illustration, that the same 0.18 dB/km main-fibre attenuation and the same connector/splice/entry losses apply, total loss = 13.3 + 1.35 = 14.65 dB.

Received power = +3 − 14.65 = −11.65 dBm. Margin = (−11.65) − (−20) = 8.35 dB.

This illustrative result exceeds 6 dB, but it is not a validated design because the actual G.655 attenuation, dispersion slope, DCF characteristics, and system tolerances must be confirmed.

8. Final Engineering Decision

Decision: NO — do not deploy the original configuration without redesign and verification.

Reason: The basic power budget without DCF passes the 6 dB margin requirement, and the calculated OSNR values exceed the 17 dB target. However, adding 10.2 km of DCF to compensate the G.652.D span raises the estimated total loss and reduces the power margin to 4.6 dB, below the required 6 dB. The simplified dispersion calculations also indicate that dispersion must be addressed. A lower-dispersion fibre such as G.655 may reduce the required compensation, but the proposed alternative remains a candidate rather than a deployment-ready design.

8.1 Technical Justification

8.1.1 Fibre and Modal Suitability

The normalised frequency of Fibre A is V ≈ 1.95. Since this is below the single-mode cutoff value of 2.405, Fibre A operates as a single-mode fibre at 1550 nm. Fibre B has V ≈ 21.30, so it is multimode and is not the preferred choice for this 60 km high-data-rate link because of its potential for intermodal dispersion.

8.1.2 Propagation and Bandwidth-Distance Performance

The required transmission distance is 60 km, with an initial data rate of 10 Gb/s and a possible future upgrade to 100 Gb/s. Single-mode propagation is appropriate because it avoids the modal dispersion associated with multimode fibre.

Chromatic dispersion can broaden pulses and cause neighbouring bits to overlap. The lower dispersion value used for G.655 in this assignment gives less pulse broadening than G.652.D. Actual system performance must be verified using the selected transmitter spectral width, receiver specifications, and permitted residual dispersion.

8.1.3 Total Optical Loss and Received Power

The total fibre attenuation is calculated by adding the supplied attenuation components:

$\alpha_{total} = 0.04 + 0.11 + 0.03 = 0.18$ dB/km

The fibre loss over 60 km is:

$L_{fibre} = \alpha_{total} \times L = 0.18$ dB/km $\times$ 60 km $= 10.8$ dB

For the G.652.D entry configuration, the total loss is:

$L_{total} = L_{fibre} + L_{connectors} + L_{splices} + L_{entry}$

$L_{total} = 10.8 + 1.6 + 0.8 + 0.8 = 14.0$ dB

Received power is transmitter power minus total link loss:

$P_R = P_T - L_{total}$

$P_R = +3$ dBm $- 14.0$ dB $= -11.0$ dBm

Power margin is received power minus receiver sensitivity:

$M = P_R - P_{sensitivity}$

$M = (-11.0$ dBm$) - (-20$ dBm$) = 9.0$ dB

The calculated 9.0 dB margin exceeds the required 6 dB by 3.0 dB.

Using the G.657.A entry-loss assumption of 0.1 dB, the total loss becomes:

$L_{total} = 10.8 + 1.6 + 0.8 + 0.1 = 13.3$ dB

$P_R = +3$ dBm $- 13.3$ dB $= -10.3$ dBm

$M = (-10.3$ dBm$) - (-20$ dBm$) = 9.7$ dB

This configuration meets the stated power-margin requirement before dispersion compensation is added.

8.1.4 Dominant Loss Mechanism and Practical Optimisation

Of the supplied attenuation components, the 0.11 dB/km component is the largest contributor to the stated fibre attenuation. The actual physical loss mechanisms must be confirmed from the assignment definitions and the fibre datasheet.

Practical improvements include using low-loss connectors and splices, cleaning and inspecting connector end faces, avoiding unnecessary connection points, and controlling fibre bending. Using G.657.A at the building entry is beneficial where bend-related loss is a concern.

8.1.5 Chromatic Dispersion, PMD, and Compensation

Chromatic pulse broadening is estimated using the magnitude of the dispersion parameter:

$\Delta\tau = |D| \times \Delta\lambda \times L$

For G.652.D, D = 17 ps/(nm·km), $\Delta\lambda$ = 0.4 nm, and L = 60 km:

$\Delta\tau = 17\ \text{ps/(nm·km)} \times 0.4\ \text{nm} \times 60\ \text{km} = 408\ \text{ps}$

For G.655, using D = 4.5 ps/(nm·km):

$\Delta\tau = 4.5\ \text{ps/(nm·km)} \times 0.4\ \text{nm} \times 60\ \text{km} = 108\ \text{ps}$

The bit period at 10 Gb/s is:

$T_b = 1 / R_b = 1 / (10 \times 10^9\ \text{s}^{-1}) = 100\ \text{ps}$

Both broadening estimates exceed one bit period under this simplified calculation. This indicates that dispersion management requires further investigation using the actual transceiver tolerance and system design criteria.

For G.652.D, the estimated DCF length is calculated by balancing the accumulated dispersion of the main span against the DCF dispersion:

$L_{DCF} = -(D_{main} \times L_{main}) / D_{DCF}$

$L_{DCF} = -(17\ \text{ps/(nm·km)} \times 60\ \text{km}) / (-100\ \text{ps/(nm·km)}) = 10.2\ \text{km}$

At a DCF attenuation of 0.5 dB/km, the added DCF loss is:

$L_{DCF_loss} = 10.2\ \text{km} \times 0.5\ \text{dB/km} = 5.1\ \text{dB}$

Using the G.657.A entry configuration as the baseline, the revised total loss is:

$L_{total,new} = 13.3\ \text{dB} + 5.1\ \text{dB} = 18.4\ \text{dB}$

$P_R = +3\ \text{dBm} - 18.4\ \text{dB} = -15.4\ \text{dBm}$

$M = (-15.4\ \text{dBm}) - (-20\ \text{dBm}) = 4.6\ \text{dB}$

This is 1.4 dB below the required 6 dB margin. Therefore, this particular DCF configuration does not satisfy the power-budget requirement without further changes.

The estimated PMD is:

$PMD \approx D_PMD \times \sqrt{L}$

$PMD \approx 0.1\ ps/\sqrt{km} \times \sqrt{(60\ km)} \approx 0.775\ ps$

This is below the simplified limits of 10 ps for 10 Gb/s and 1 ps for 100 Gb/s used in the assignment analysis. However, the 100 Gb/s upgrade requires confirmation against the selected equipment PMD tolerance and complete system specifications.

8.1.6 OSNR and Amplification Strategy

Before amplification, the calculated OSNR is signal power minus ASE noise power, provided both powers are measured in the same reference bandwidth:

$OSNR = P_signal - P_ASE$

$OSNR = (-16\ dBm) - (-39\ dBm) = 23\ dB$

This is 6 dB above the required 17 dB.

After the assumed amplifier gain and noise increase, the signal power is −2 dBm and the ASE noise power is −23 dBm:

$P_signal,after = -16\ dBm + 14\ dB = -2\ dBm$

$P_ASE,after = -39\ dBm + 16\ dB = -23\ dBm$

$OSNR_after = (-2\ dBm) - (-23\ dBm) = 21\ dB$

The resulting OSNR is 4 dB above the required target. Although this meets the assignment OSNR requirement, the amplifier reduces OSNR by 2 dB in the given calculation. I recommend a low-noise amplifier with appropriately selected gain, verified noise figure, and OSNR monitoring. Amplification must be based on the complete link budget rather than used to compensate for an unresolved design problem.

8.2. Proposed Redesign

Fibre configuration: Use single-mode fibre for the 60 km main span. Evaluate G.655 as a lower-dispersion alternative and use G.657.A at the building entry if bend-insensitive fibre is required.

Link-loss management: Retain low-loss connectors and splices, minimise unnecessary connection points, and verify actual attenuation values for the selected fibre.

Dispersion management: Recalculate the compensation requirement using the selected fibre and actual transmitter/receiver specifications. Avoid installing 10.2 km of DCF without addressing its additional 5.1 dB loss.

DCF alternative: If the G.655 dispersion value of 4.5 ps/(nm·km) is confirmed, the illustrative compensation length is $L_{DCF} = -(4.5 \times 60)/(-100) = 2.7$ km. At 0.5 dB/km, this contributes 1.35 dB of loss.

Illustrative power budget: Assuming the same baseline link losses, total loss = 13.3 + 1.35 = 14.65 dB; received power = +3 − 14.65 = −11.65 dBm; margin = −11.65 − (−20) = 8.35 dB. This is provisional because actual G.655 attenuation and dispersion characteristics have not been confirmed.

Amplifier and OSNR strategy: Use an amplifier only if the verified power budget requires it. Select gain and noise figure to preserve the required OSNR and avoid unnecessary amplification.

Commissioning and verification: Measure insertion loss and OSNR, verify residual dispersion and PMD against selected equipment specifications, and perform bit-error-rate (BER) testing before approving service.

8.3. Predicted Effect of the Redesign

The proposed redesign is expected to reduce chromatic dispersion and avoid the excessive loss associated with the original 10.2 km DCF proposal. Under the stated assumptions, the illustrative G.655 configuration with 2.7 km of DCF provides a power margin of approximately 8.35 dB, compared with 4.6 dB for the G.652.D-plus-10.2 km DCF case.

This is an improvement in the calculated power budget, but it does not by itself prove that the link is deployment-ready. Actual fibre attenuation, residual dispersion, amplifier noise, equipment compatibility, and the 100 Gb/s upgrade requirements must still be checked.

Conclusion

The design study shows why an optical link cannot be approved using received power alone. Fibre type, connector and splice losses, dispersion, PMD, amplifier noise, and the future data-rate target must all be checked together. The basic 60 km link has a satisfactory estimated power margin, but the G.652.D-plus-DCF case fails the stated 6 dB margin. The recommended engineering action is to revise the dispersion-management approach and verify the complete design using actual component specifications before deployment.